DAVID PETRIE

Systems & Security | Distributed Infrastructure | Penn State Class of 2027
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SUMMARY

Enterprise Technology Integration student at Penn State who builds and runs real systems: a multi-region Solana and Aptos RPC network that peaked at 8M+ transactions in a day, a live SMS product, and a sports-market pricing engine in Go. Seeking internships and full-time entry-level roles in systems, infrastructure, and security.

TECHNICAL SKILLS

Programming: Python, Go, TypeScript, JavaScript, Rust, SQL
Infrastructure: Linux, Docker, Caddy, PostgreSQL, SQLite, TimescaleDB, Azure, Git, CI/CD, Server Orchestration
Security: IDS/SIEM, Snort, Wireshark, Nmap, Nessus, Hydra, Vulnerability Assessment
Blockchain: RPC Design, Solana, Aptos, Yellowstone gRPC, Node Deployment
AI/ML: TensorFlow, PyTorch

EXPERIENCE

PixelLabz (formerly PixelPilotz) — Co-Founder & Director of Community Relations | 2022–2025

- Co-architected a non-consensus RPC network on Solana and Aptos, raising ~$180,000 through token sales.
- Ran 15+ bare-metal servers in the U.S., Germany, and Singapore at 98% uptime; 8M+ transactions on peak day.
- Directed growth to 4,000+ users; delivered demos, onboarding, and technical support.

NodeStop LLC — Information Security Analyst, Jr (Internship) | 2023–2025

- Conducted vulnerability assessments and risk analysis for enterprise systems.
- Supported compliance readiness and remediation planning.
- Assisted with IDS/SIEM implementation; communicated findings to technical and non-technical teams.

PROJECTS

Ringlock — Missed-Call Text-Back for Contractors | ringlock.tech | 2026

- Built and launched a multi-tenant SMS product that texts missed callers back in ~3 seconds, asks for the job address, and forwards replies to the contractor.
- Runs on Docker, PostgreSQL, n8n, Twilio, and Caddy; verifies Twilio signatures and handles STOP opt-outs.

Kalshi Edge-Finder — Sports Market Pricing Engine in Go | 2026

- Prices Kalshi sports markets from sharp sportsbook lines and an Elo model, with out-of-sample backtests.
- Keeps real money off until a sport clears backtest and closing-line gates; uses fractional Kelly sizing.

Portfolio Site — Interactive 3D Portfolio | davelabs.me | 2026

- Wrote a custom sphere-collision physics engine on Three.js with a cursor-driven collider and reduced-motion fallback.
- Framework-free static site on Vercel with a strict Content Security Policy and HSTS.

EDUCATION

The Pennsylvania State University — B.S. Enterprise Technology Integration | **Expected 2027**
GPA: 3.4
Relevant coursework: Systems Architecture, AI & Machine Learning, Distributed Systems & Blockchain, Data Analytics